19.3 Connectness.

Theorem 3.9.0 (Bertini)

Let X variety $f: X \to \mathbb{P}^r$ morphism. Fix an integer $d < \dim f(X)$

If $L \subseteq \mathbb{P}^r$ general $(r-d)$-plane, then $f^{-1}(L)$ is irreducible.

dense open in Grassmanian.

proof

Lemma. Let $p: V \to T$ dominating morphism of irre complex variety.

if p admits a section $s: T \to V$ st $s(T) \nsubseteq \mathrm{Sing}(V)$

then fibre $V_t = p^{-1}(t)$ is irreducible for general $t \in T$.

proof. $s(T) \nsubseteq \mathrm{Sing} V \Rightarrow \eta_T \notin s^{-1}(\mathrm{Sing} V)$ $\quad U := T \setminus s^{-1}(\mathrm{Sing} V)$

then shrink U with $p^{-1}(U)$ WMA T smooth $s(T) \cap \mathrm{Sing} V = \emptyset$

and p is flat, then fibre of pure dim. now consider

$V_t \cap \mathrm{Sing}(V)$, since $V_t \nsubseteq \mathrm{Sing} V$ $V_t \setminus \mathrm{Sing} V$ dense in V_t, consider $V \setminus \mathrm{Sing} V$

then WMA V smooth, then again shrink on the target

by Generically smoothness WMA p smooth.

it suffices to show fibre of p are generally connected

By [SP 0577] it suffices to show V_η geometrically irreducible

by [SP 0BUQ] it suffices to show $K(T)$ separably closed in $K(V)$.

Consider Stein factorization: take $K(T) \subseteq \widetilde{K(T)} \subseteq K(V)$ $\widetilde{K(T)}$ sep closure in $K(V)$

$\tilde{T}$ normalization of T w.r.t. $\widetilde{K(T)}$, then $V \to \tilde{T} \to T$

$\tilde{p}$ induced by $\mathcal{O}_T \to p_* \mathcal{O}_V$ since V normal

$\mathcal{O}_T \to \tilde{p}_* \mathcal{O}_{\tilde{T}} \to p_* \mathcal{O}_V$

then f finite, $\tilde{p}$ has generically connected fibre.

but $ps = id$ $\quad fps = id$ $\quad ps$ section of $f \Rightarrow \tilde{T} = T$

Now we turn to the proof of theorem.

$n := \dim f(X)$ then it suffices to prove the theorem for $d = n-1$

since if $d=n-1$ proved, for other $d<n-1$ L $(r-d)$-plane

if $f^{-1}(L)$ reducible for general L, WMA $L \supseteq L'$ L' $(r-n+1)$-plane

s.t L' is "general" i.e. $f^{-1}(L')$ irreducible.

WMA $f^{-1}(L) = Y_1 \cup Y_2$ Y_1 irreducible then $\dim Y_1 \geq n-d$ $\dim Y_2 \geq n-d$

$\exists\, k := n-1-d$ then $\dim Y_i \geq k+1$ $f^{-1}(L') = (f^{-1}(L') \cap Y_1) \cup (f^{-1}(L') \cap Y_2)$

adjust L' s.t. $f^{-1}(L') \cap Y_1 \cap Y_2 = \dim Y_1 \cap Y_2 - k$ ($\dim \geq 1$)

if $f^{-1}(L') \cap Y_1 \subseteq f^{-1}(L') \cap Y_2$ $f^{-1}(L') \cap Y_1 \subseteq f^{-1}(L') \cap Y_1 \cap Y_2$ contradiction!

($\dim \geq \dim Y_1 - 1$ $\dim \leq \dim Y_1 - 1 - k$)

same for $Y_2 \Rightarrow f^{-1}(L')$ reducible

contradiction! then $f^{-1}(L)$ irreducible.

Fix general $(r-n)$-plane $\Lambda \subseteq \mathbb{P}^r$ then set of all $(r-n+1)$-planes

$L \subseteq \mathbb{P}^r$ passing through Λ is parameterized by $T = \mathbb{P}^{n-1}$

$V := \{(x,t) \mid f(x) \in L_t\} \subseteq X \times T$ incident family $p : V \to T$

then it suffices to show irreducibility of general fibre

Now observe $V \to X$, $\forall x \in X \setminus f^{-1}(\Lambda)$ fix x and Λ defines unique t

i.e. $V \setminus p^{-1}(\Lambda) \times T \to X \setminus f^{-1}(\Lambda)$ $V \setminus p^{-1}(\Lambda) \times T$ is dense in V

then V irreducible, and note that $\mathrm{Bl}_\Lambda \mathbb{P}^r \hookrightarrow \mathbb{P}^r \times T$

the image is $\{(x, L_t) \mid x \in L_t\}$ then $\mathrm{Bl}_{f(X)} X \hookrightarrow X \times T \to X$

(intersect properly)

(strict trans) $\mathrm{Bl}_\Lambda \mathbb{P}^r \hookrightarrow \mathbb{P}^r \times T \to \mathbb{P}^r$

choose Λ s.t $f^{-1}(\Lambda)$ cut by regular sequence

then $V = \mathrm{Bl}_{f(X)} X$ i.e. $\forall x \in f^{-1}(\Lambda) \cap X^{reg}$ V smooth near x.

now $X \to \overline{f(X)}$ generically smooth on the source

then $\exists U \subseteq X$ $V \subseteq f(X)$ $U \to V$ smooth, $U \subseteq X^{reg}$

choose Λ s.t. $\Lambda \cap V \neq \emptyset$ then $\exists$ such $q \in f^{-1}(\Lambda) \cap X^{reg}$ also in $f^{-1}(\Lambda)^{reg}$

$T \to V$ $t \mapsto (x_0, t)$ gives section of $V \to T$,

then by Lemma $S(T) \subseteq V^{reg} \Rightarrow$ theorem ✓ □

Theorem 3.9.1. (Proper version)

X proper irr variety. $\Lambda \subseteq \mathbb{P}^r$ linear subspace of codim d < $\dim f(X)$

then $f^{-1}(\Lambda)$ is connected for (all) Λ. (generally if $\exists V \supseteq \Lambda$ $f^{-1}(V) \to V$ proper)

proof G Grassmanian parametrizing all codim d linear spaces in $\mathbb{P}^r$

$Z := \{(x, \Lambda) \mid x \in f^{-1}(\Lambda)\} \subseteq X \times G$ closed Z/X Grassmanian bundle

irreducible smooth bundle + irreducible fibre $\Rightarrow Z$ irreducible

$p: Z \to G$ proper since X proper, then consider

stein factorization $Z \xrightarrow{s} W \xrightarrow{r} G$

general fibre of p is irreducible by Theorem 3.9.0.(Bertini)

then r is generically bijective, but G smooth $\Rightarrow$ normal

$[K(W) : K(G)] = 1 \Rightarrow W = G$ i.e. fibre of p is connected □

Corollary 3.9.2

X irreducible quasi-proj variety, $x_1, x_2 \in X$ distinct points, then x_1, x_2 can be joined by irreducible curve $T \subseteq X$, which may not be smooth.

proof $X' := Bl_{x_1, x_2} X$ take quasi-finite $X' \xrightarrow{f} \mathbb{P}^r$, $\Lambda \subseteq \mathbb{P}^r$ of codim $\dim X - 1$

$E_1, E_2 \subseteq X'$ exceptional divisor, then $f(E_i) \cap \Lambda \neq \emptyset$, take general Λ, then

$f|_{E_i}$ finite.

by Theorem 3.3.0. $f^{-1}(L)$ irreducible of dim 1. $T := \pi(f^{-1}(L))$ $\tilde{T} := \overline{T}$ □

Theorem 3.3.6. (Fulton & Hansen)

X irreducible proper variety, $f: X \to \mathbb{P}^r \times \mathbb{P}^r$, assume $\dim f(X) > r$.

then $f^{-1}(\Delta) \subseteq X$ is connected $(\Delta: \mathbb{P}^r \hookrightarrow \mathbb{P}^r \times \mathbb{P}^r)$

proof. Observation: $U \hookrightarrow \mathbb{P}^{2r+1}$ where $U = \mathbb{P}^{2r+1} \setminus (L_1 \cup L_2)$ (proj subspace.)

$\downarrow$

$\mathbb{P}^r \times \mathbb{P}^r$

$L_1 = \{(0, \dots, 0, y_0, \dots, y_r)\}$ $L_2 = \{(x_0, \dots, x_r, 0, \dots, 0)\}$

then $U \to \mathbb{P}^r \times \mathbb{P}^r$ defined by $[x_0, \dots, x_r, y_0, \dots, y_r] \mapsto ([x_0, \dots, x_r], [y_0, \dots, y_r])$

Locally on $\mathbb{P}^r \times D(y_i)$, $U|_{(\mathbb{P}^r \times D(y_i))} \hookrightarrow \mathbb{A}^{2r+1} \setminus (0 \times \mathbb{A}^r)$

$\downarrow$ $\downarrow$ $(\mathbb{A}^{r+1} \setminus 0) \times \mathbb{A}^r$

$\mathbb{P}^r \times D(y_i) \hookrightarrow \mathbb{P}^r \times \mathbb{A}^r$

where $\mathbb{A}^{r+1} \setminus 0 \to \mathbb{P}^r$ natural proj.

then $U / \mathbb{P}^r \times \mathbb{P}^r$ is principal G_m-bundle.

$L \subseteq U$ proj subspace $\{[x_0, \dots, x_r, y_0, \dots, y_r] \mid x_i = y_i\}$.

One can check that $L \hookrightarrow \Delta$, now

$f \circlearrowright f$

$U \to \mathbb{P}^r \times \mathbb{P}^r$

Given $X \xrightarrow{f} \mathbb{P}^r \times \mathbb{P}^r$

$V_X \xrightarrow{f_r} U$

$\downarrow$ □ $\downarrow$

$X \xrightarrow{f} \mathbb{P}^r \times \mathbb{P}^r$

X irre + V_X / X smooth with irre fibre

$\Rightarrow V_X$ irre (V_X / X open map)

f proper $\Rightarrow f_r$ proper.

then $V_X \xrightarrow{f_r} U \hookrightarrow \mathbb{P}^{2r+1}$ $\dim f_r(V_X) = \dim \overline{f_r(V_X)}$

$V_X \to V_{f(X)} \hookrightarrow U$

$\downarrow$ □ $\downarrow$ □ $\downarrow$

$X \to f(X) \hookrightarrow \mathbb{P}^r \times \mathbb{P}^r$

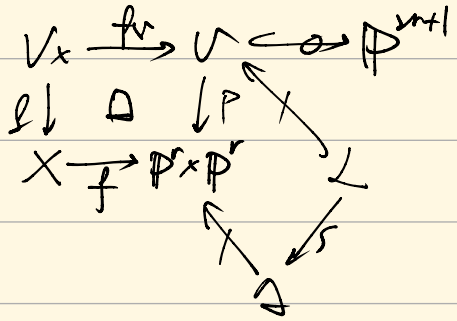
then $V_{f(X)} = \mathrm{Im} f_r$ G_m-bundle / $f(X)$

then $\dim V_{f(X)} = \dim f(X) + 1$ (flatness)

we need connectness

$\dim \overline{f_V(V_X)} > r+1 = \operatorname{codim} L$ then by Thm 3.3.3. $f_V^{-1}(L)$ connected

set-theoretically $\rho(f_V^{-1}(L)) = f^{-1}(\Delta)$ connected.



Check: first $\rho(f_V^{-1}(L)) \subseteq f^{-1}(\Delta)$

conversely $\forall x \in f^{-1}(\Delta)$ lift $f(x)$ to $y \in L$

then $\exists!\ (x,y) \in V_X$ $f_V(x,y) = y$ $\rho(x,y) = x$

$x \in \rho(f_V^{-1}(L))$ since $(x,y) \in f_V^{-1}(L)$ □

Example 3.3.7

$X, Y \subseteq \mathbb{P}^r$ be irreducible subvarieties, if $\dim X + \dim Y > r$ then $X \cap Y$ connected.

since $X \cap Y$ is preimage of Δ along $X \times Y \hookrightarrow \mathbb{P}^r \times \mathbb{P}^r$

$\dim X \times Y = \dim X + \dim Y$ $(k = \bar{k})$ $\Rightarrow \dim X \times Y > r$.

Example 3.3.8.

Fact: $f: X \to \mathbb{P}^r \times \cdots \times \mathbb{P}^r$, X irre proj $\dim f(X) > (k-1)r$

then $f^{-1}(\Delta)$ is connected $(V \hookrightarrow \mathbb{P}^{kr+k-1})$

* Grothendieck Connectedness Theorem

Def. 3.3.13. (Connectedness in dim k)

A Noetherian scheme X is connected in dim k if $\dim X > k$ and

$X \setminus T$ is connected $\forall T \subseteq X$ of $\dim < k$.

Example 3.3.14.

① X connected $\iff$ X 0-connected $(\dim\emptyset = -1)$

② If X d-connected, $d' \le d \Rightarrow X$ is d'-connected

③ If X irre of dim n then X is $(n-1)$-connected

(X irre $\iff$ X is "$\dim X$-connected")

Example. 3.3.15.

$\forall k < \dim X$, X is k-connected $\iff$ $\forall$ irre comp $X' \neq X''$

sequence of components s.t $\dim(X_{i-1} \cap X_i) \ge k$. where $X_0 = X'$ $X_\ell = X''$ $X_i \neq X_j$

Q: Why we need $\dim Y < k < \dim X$ what if $\dim Y = \dim X - 1$?

If $X = X_1 \cup \cdots X_n$ union of irre components.

assume $\dim X_1 \ge \dim X_2 \ge \cdots \ge \dim X_n$ then $\dim X_1 \cap X_i \le \dim X_1 - 1 = \dim X - 1$

but $\left(X_1 \setminus \bigcup_{i\ge 2} X_i\right) \cap \left(\bigcup_{i\ge 2} X_i \setminus X_1\right) = \emptyset \Rightarrow X \setminus \bigcup (X_1 \cap X_i)$ cannot be connected.

$\operatorname{codim} \ge 1$

then one need to start with $\dim Y \le \dim X - 2$.

Theorem 3.3.16 (Grothendieck Connectedness Thm)

Let (A, m) be a complete local Noetherian ring, $k \ge 1$.

$X = \operatorname{Spec} A$ k-connected $\underset{\text{flem}}{\Rightarrow}$ $\operatorname{Spec} A/fA$ is $(k-1)$-connected.

Corollary. 3.3.17.

(A, m) be a complete local Noetherian domain of dim n.

for $d \le n-1$ $f_1, \dots, f_d \in m$, then $\operatorname{Spec}(A/(f_1, \dots, f_d))$ is $(n-d-1)$-connected

Corollary. 3.3.18 (Stronger Bertini)

$X \subseteq \mathbb{P}^r$ closed alg subset k-connected. $L \subseteq \mathbb{P}^r$ subspace codim d.

then $X \cap L$ is connected in dim $k-d$.

In particular if X irreducible $k = n-1$, if $d < n$ then $k-d \ge 0$, $X \cap L$ connected.

Example. 3.3.19. (Verify Set-theo LCI)

$X \subseteq \mathbb{A}^6_{\mathbb{C}}$ $X = V(x_1, x_2, x_3) \cup V(x_4, x_5, x_6) = V(x_1x_4, x_1x_5, x_1x_6, x_2x_4, x_2x_5, x_2x_6, x_3x_4, x_3x_5, x_3x_6)$

$\dim X = 3$ Q: is X set-theo LCI (check on stalk/completion) by 3 functions at 0?

$V(x_1, x_2, x_3) \cap V(x_4, x_5, x_6) = 0 \Rightarrow X$ is 0-connected but not 1-connected.

if $U \ni 0$ affine integral, $U = \operatorname{Spec} A$ if $U \cap X = \operatorname{Spec} A/(f_1, f_2, f_3)$

then $U \cap X$ is $(6-1-3=2)$-connected X

even wrong for $A/(f_1 f_2 f_3 f_4)$, What about 5?

Answer: X is set theo CI of 3 functions.

Consider abstract polynomial $x_1 + x_2 T + x_3 T^2 =: f_1$ $x_4 + x_5 T + x_6 T^2 =: f_2$

then $f_1 f_2 = 0$ iff $f_1 = 0$ or $f_2 = 0$ iff $x \in X$.

Compute coefficient of $f_1 f_2$: $f_1 f_2 = x_1x_4 + (x_1x_5 + x_2x_4)T + (x_1x_6 + x_3x_4 + x_2x_5)T^2$

$+ (x_2x_6 + x_3x_5)T^3 + x_3x_6 T^4$

Theorem 3.3.20 (Ramification)

X complete irr variety of dim n, $f: X \to \mathbb{P}^r$ unramified

if $2n > r$ then f is closed embedding.

($2n > r$ is sharp since if $n=1$ $r=2$ $X = \mathbb{P}^1$ then

$X \to C \to \mathbb{P}^2$ where C nodal curve, $X - \tilde{C} \to C$ unramified)

* One has to fold something to map X into $\mathbb{P}^r$ if $2n > r$.

proof fact: inj on k-pt + unramified $\Rightarrow$ closed immersion

consider $F = f \times f : X \times X \to \mathbb{P}^r \times \mathbb{P}^r$ then $F^{-1}(\Delta_{\mathbb{P}^r}) = \{(x,y) \mid f(x) = f(y)\} \subseteq X \times X$

$f(k)$ inj iff $F^{-1}(\Delta_{\mathbb{P}^r}) = \Delta_X$ ($\forall x \neq y$ $f(x) \neq f(y)$).

$F^{-1}(\Delta_{\mathbb{P}^r}) = X \times_{\mathbb{P}^r} X$. connected by Theorem 3.3.6 ($\dim \text{Inf} = \dim X \times X = 2n > r$)

unramified + proper $\Rightarrow$ finite.

X/Y separated + unramified $\Rightarrow$ $\Delta_{X/Y}: X \to X \times_Y X$ is clopen immersion

then $\Delta_X \subseteq X \times_{\mathbb{P}^r} X$ clopen $\Rightarrow$ $\Delta_X = X \times_{\mathbb{P}^r} X$. i.e. f inj. □

Example 3.9.2

If $r \geq 3$, there are no irr singular hypersurfaces $X \subseteq \mathbb{P}^r$ with only normal crossing singularities.

* Definition of normal crossing singularity

X has normal crossing singularity at p iff $\widehat{\mathcal{O}}_{X,x} \to k[[x_1 - x_n]] / (x_1 \cdots x_r)$

iff étale neighborhood (V, v) $U = \operatorname{Spec} k[x_1 \cdots x_n] / (x_1 \cdots x_r)$

(Artin approximation) Alper B.5.2

$(X, p) \leftarrow (V, v) \rightarrow (U, 0)$

iff analytically $\exists U \ni p$ $U \hookrightarrow V \cap V(z_1 \cdots z_r) \subseteq \mathbb{C}^n$ $V \subseteq \mathbb{C}^n$ open.

(Cartan-Cartan extension)

If X has only normal crossing singularities, then at singular pts

$\widehat{\mathcal{O}}_{X,a} = \widehat{\mathcal{O}}_{X,x} = k[[x_1 \cdots x_n]]/(x_1 \cdots x_r)$ $\quad k[[x_1 \cdots x_n]]/(x_1 \cdots x_r) \hookrightarrow k[[x_1 \cdots x_n]]/x_1 \times \cdots \times k[[x_1 \cdots x_n]]/x_r$

$= k[[x_1 \cdots x_n]]^r$

check tangent space $T_x\check{X} = T(\mathcal{O}_{X,x}\check{)} = T(\widehat{\mathcal{O}}_{X,x}\check{)} = (x_1, \cdots x_n)/(x_1 \cdots x_r) + (x_i x_j)$

$(\text{WMA } r \geq 2) = (x_1, \cdots x_n)/(x_i x_j) = \{\sum a_i x_i \mid a_i \in k\}$.

normalization commutes with localization $\Rightarrow$ $\pi\mathcal{O}_{\tilde{X},y_i} \longrightarrow \tilde{X}$

$\square$

$\mathcal{O}_{X,x} \longrightarrow X$

analytically $\widehat{\mathcal{O}}_{X,x} = \pi\mathcal{O}_{\tilde{X},y_i} \xrightarrow{\text{match}} \widehat{\mathcal{O}}_{\tilde{X},y_i} = k[[x_1 \cdots x_n]]$

tangent space k^{n-1}, tangent map $T_x\check{} \longrightarrow T_{y_i}\check{}$

$\sum a_i x_i \longmapsto (a_1 \cdots \hat{a}_i \cdots a_n)$

$\tilde{X} \to X$ unramified. $\tilde{X}$ regular.

Now we show If $r \geq 3$, there are no irre singular hypersurfaces $X \subseteq \mathbb{P}^r$ with only normal crossing singularities.

if so, $\tilde{X} \to X \to \mathbb{P}^r$ $\quad \dim \tilde{X} = r-1$ $\quad 2(r-1) > r$

then $\tilde{X} \to \mathbb{P}^r$ unramified $\Rightarrow \tilde{X} \hookrightarrow \mathbb{P}^r \Rightarrow X$ regular contradiction!

Example. 3.93.

$X \subseteq \mathbb{P}^r$ irre subvariety of $\dim n$, $2n > r$, then $\pi_1^{\text{ét}}(X) = e$.

proof Suppose $p: Y \to X$ connected finite étale cover

$Y' \subseteq Y$ irreducible component of Y, then $Y' \hookrightarrow Y \to X \hookrightarrow \mathbb{P}^r$ unramified

then $Y' \hookrightarrow \mathbb{P}^r \Rightarrow Y' \hookrightarrow X$ $\square$